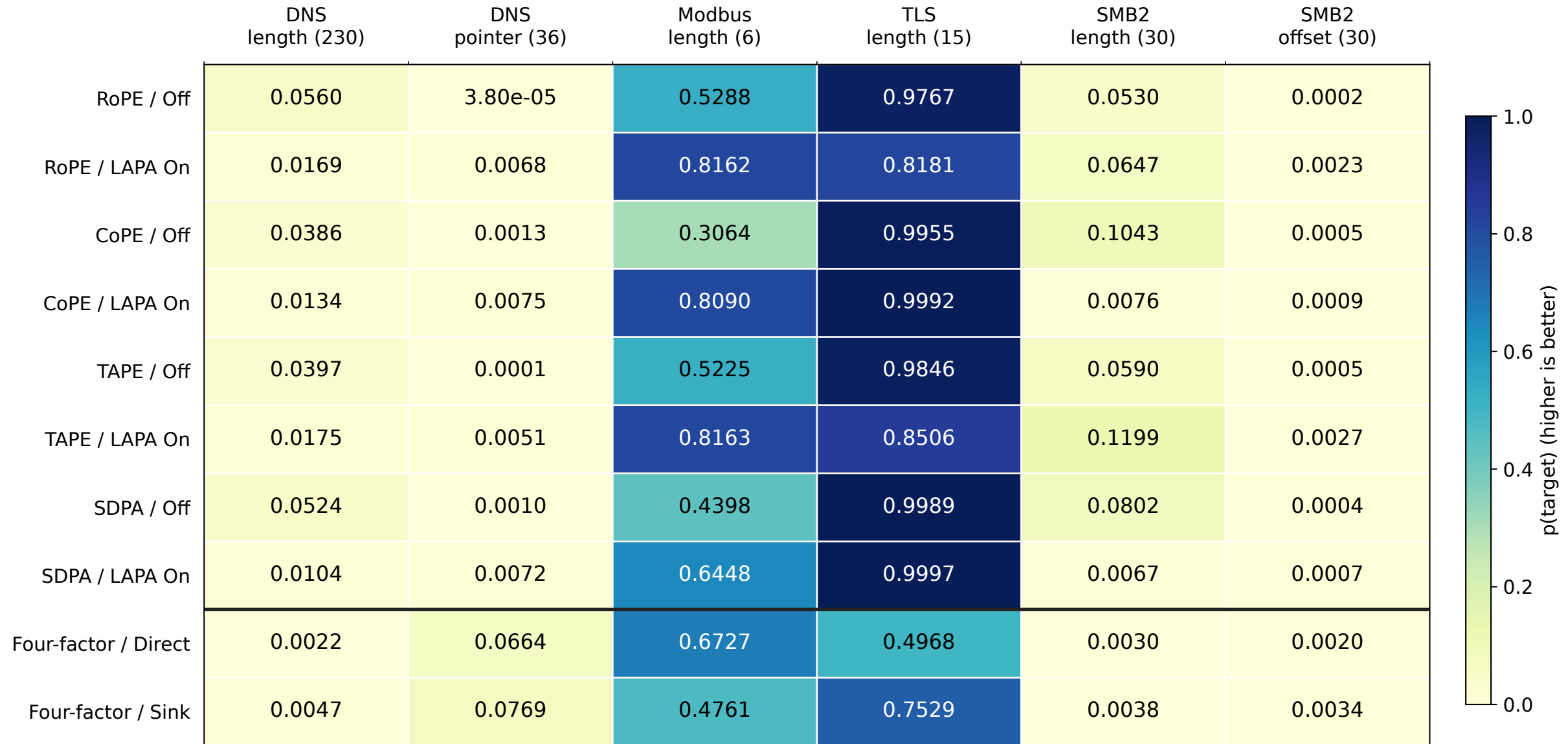


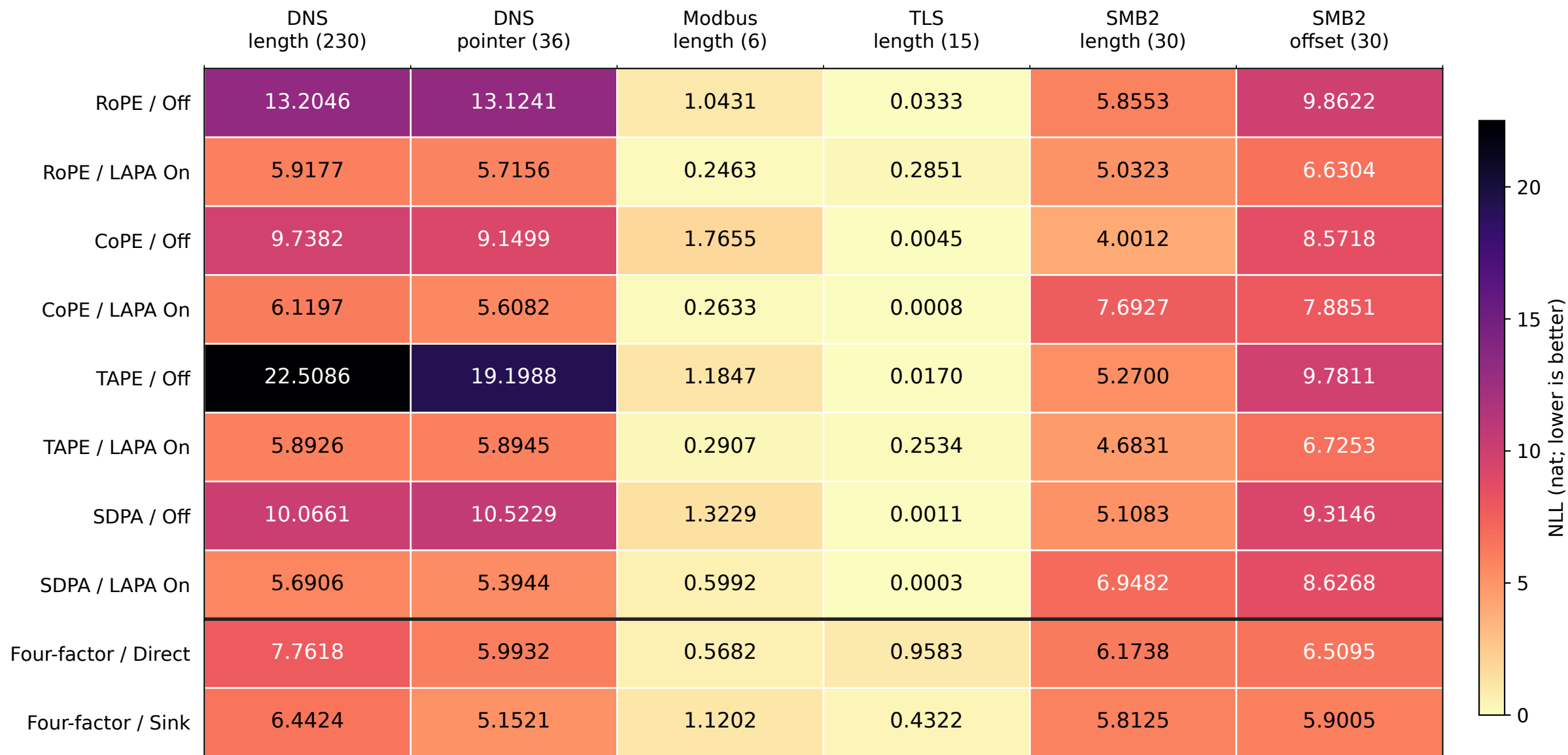
Correct-destination probability | same held-out fields

New rows use Source / Width / Endian / Base in every layer; no QK similarity. Learned V is retained.



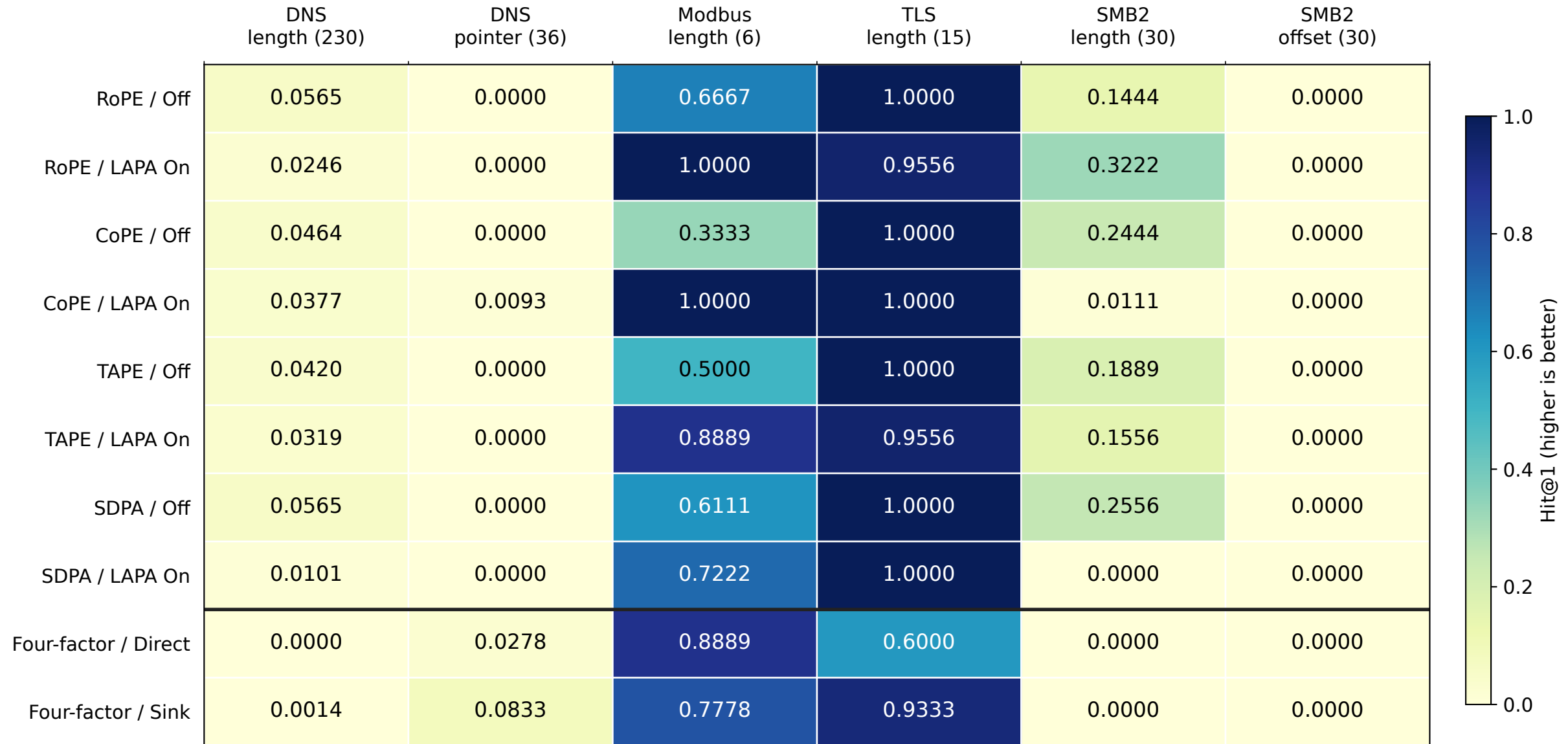
Correct-destination NLL | same held-out fields

New rows use Source / Width / Endian / Base in every layer; no QK similarity. Learned V is retained.



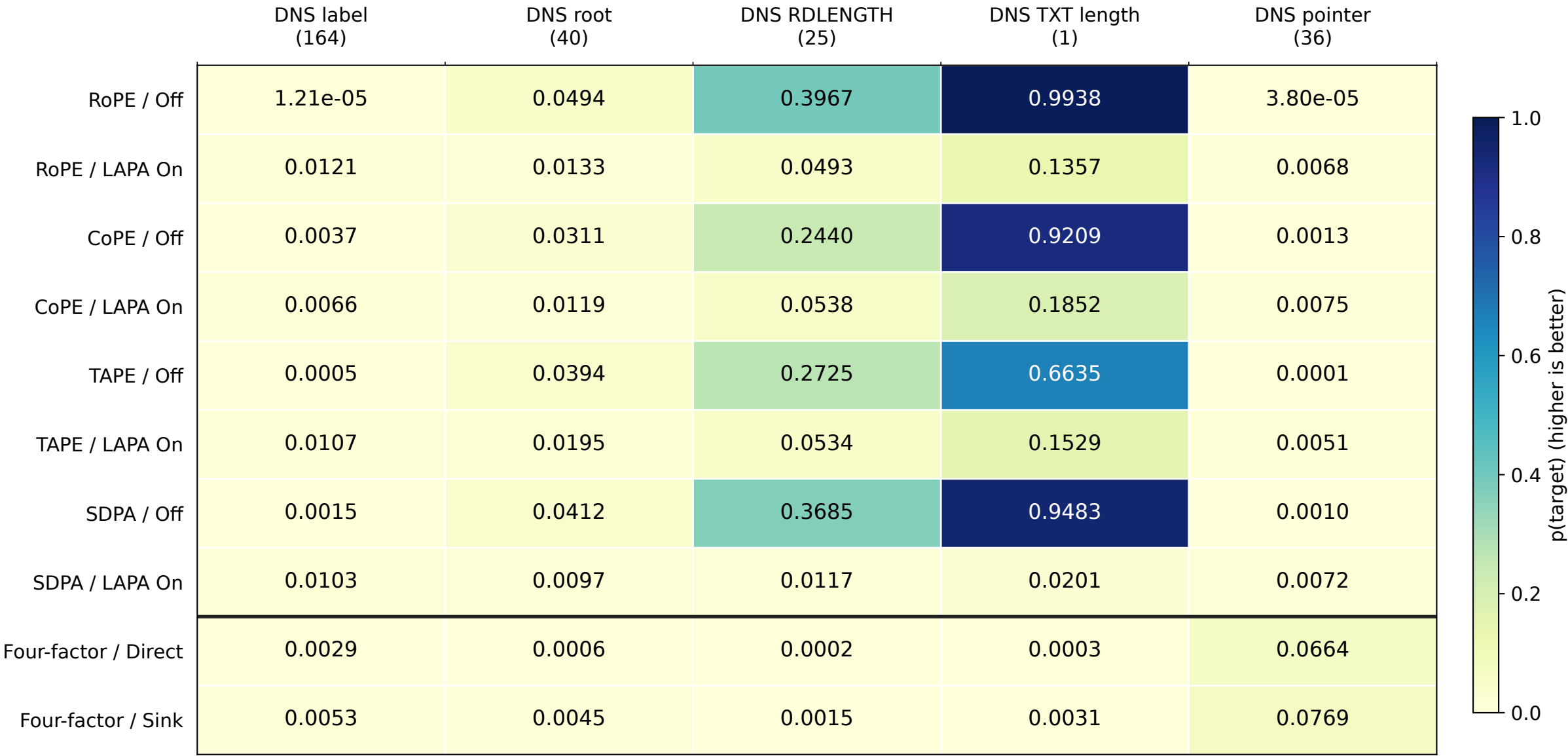
Correct-destination Hit@1 | same held-out fields

New rows use Source / Width / Endian / Base in every layer; no QK similarity. Learned V is retained.



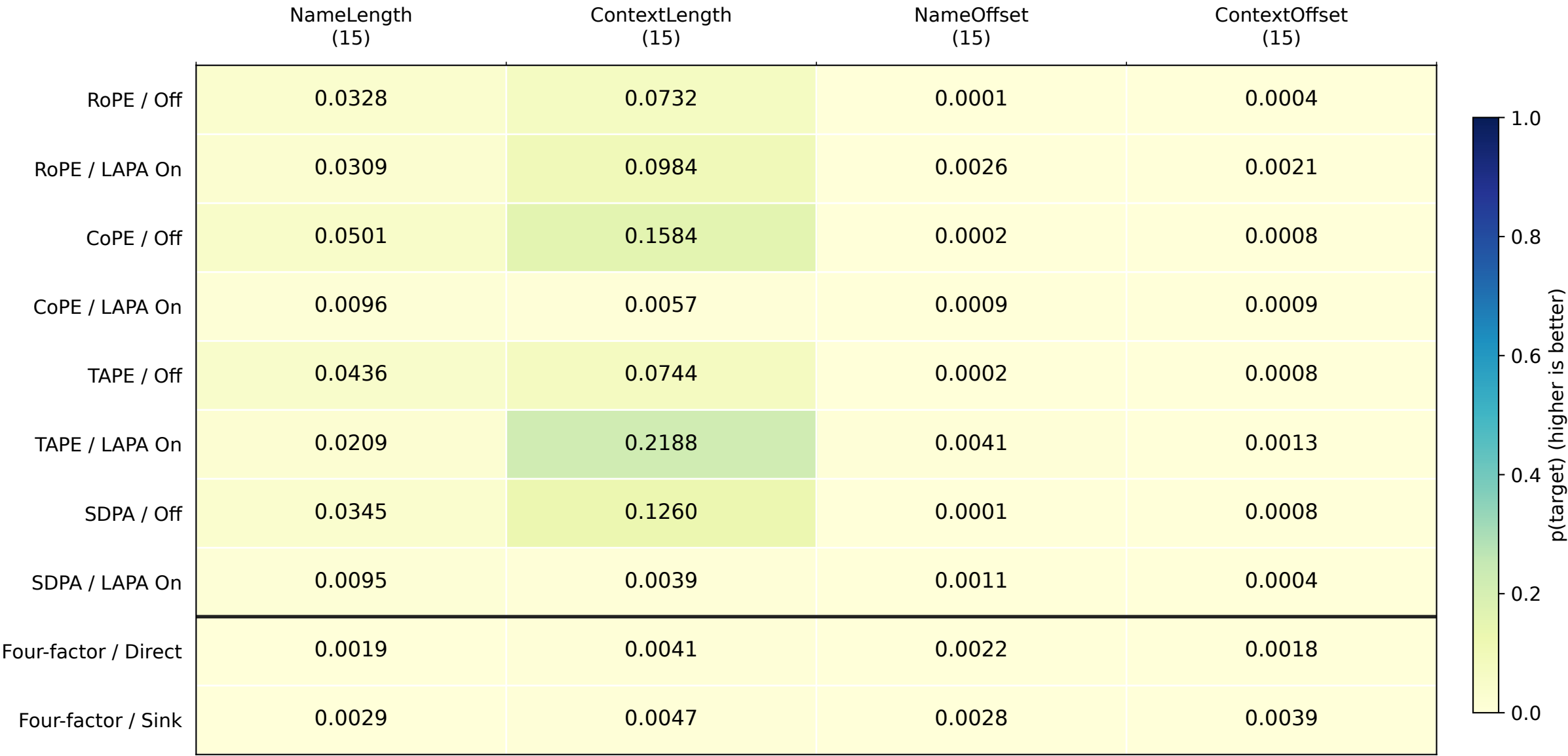
DNS detail | length subtypes and pointer

New rows use Source / Width / Endian / Base in every layer; no QK similarity. Learned V is retained.



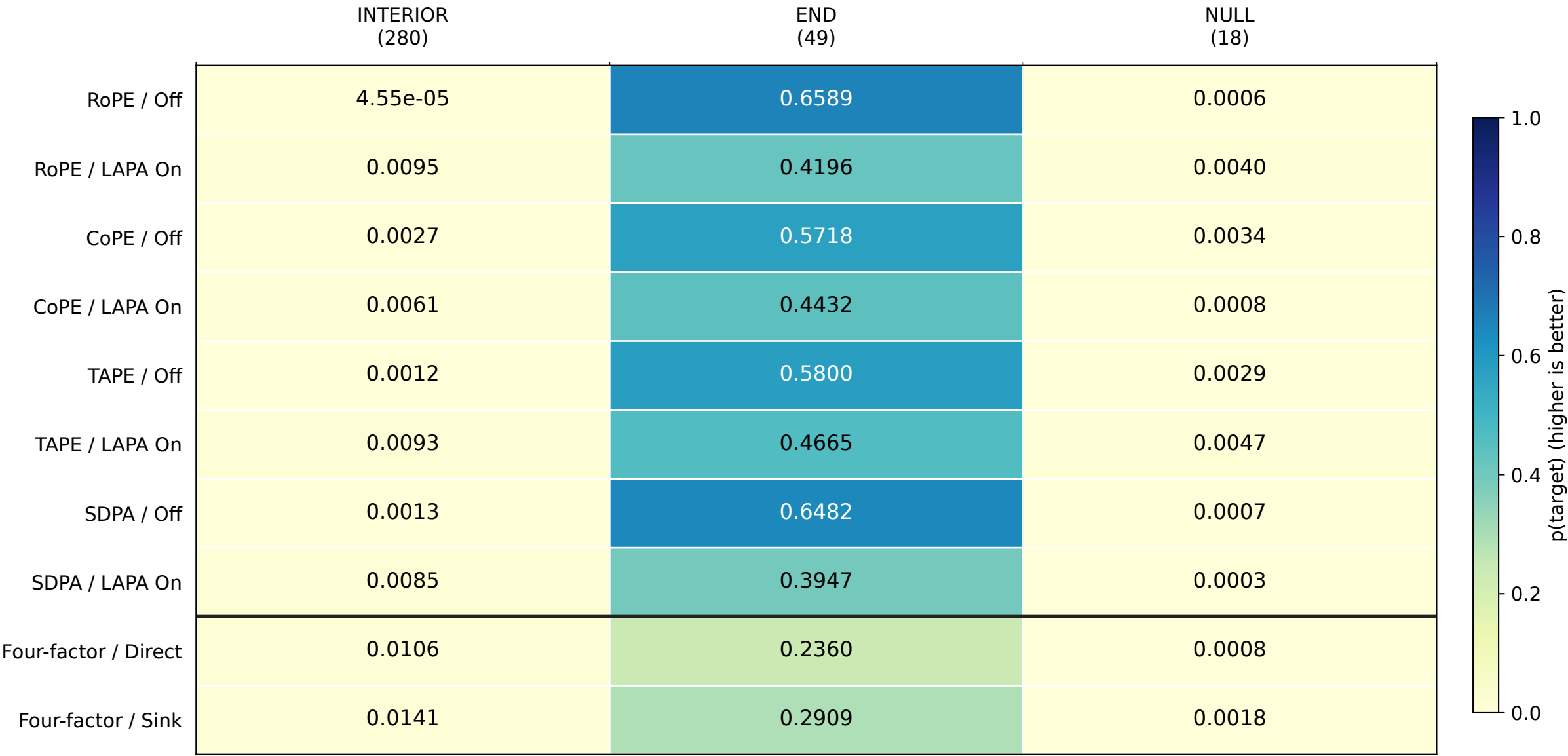
SMB2 detail | length and offset

New rows use Source / Width / Endian / Base in every layer; no QK similarity. Learned V is retained.



Endpoint strata | existing fields only

New rows use Source / Width / Endian / Base in every layer; no QK similarity. Learned V is retained.



Four learned factors | post-inference diagnostics

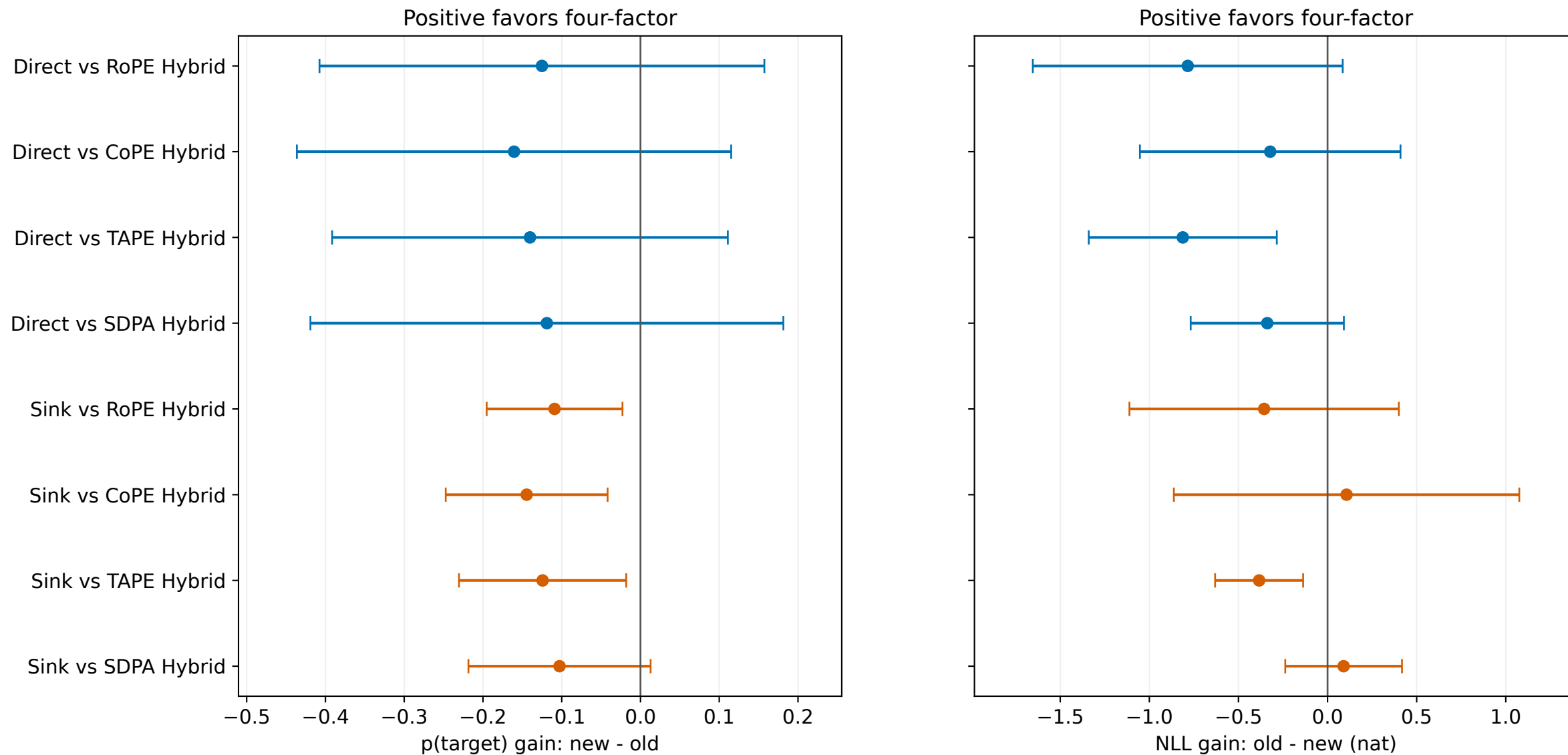
Gold source is used for indexing only after predictions are sealed; it is never a forward input.

	Source p	Width p true source	Endian p (equiv.) true source	Base p true source	Valid mass
Direct / DNS LENGTH	0.0044	0.0915	0.9092	0.0699	0.6690
Direct / DNS POINTER	0.0002	0.2672	0.2794	0.1072	0.7149
Direct / modbus LENGTH	0.0340	0.2733	0.8859	0.8326	0.5617
Direct / tls LENGTH	0.3918	0.8244	0.9960	0.9914	0.6559
Direct / SMB2 LENGTH	0.0034	0.1886	0.0605	0.0401	0.7690
Direct / SMB2 OFFSET	0.0010	0.1543	0.1229	0.0462	0.6737
Sink / DNS LENGTH	0.0017	0.1269	0.8993	0.0813	0.8603
Sink / DNS POINTER	0.0002	0.3610	0.2623	0.0669	0.8497
Sink / modbus LENGTH	0.0262	0.5420	0.4215	0.4853	0.5794
Sink / tls LENGTH	0.7027	0.9844	0.9914	0.9843	0.9614
Sink / SMB2 LENGTH	0.0032	0.1038	0.0655	0.0054	0.8109
Sink / SMB2 OFFSET	0.0020	0.0665	0.1185	0.0117	0.7827

Width-1 endian equivalents are summed. Valid mass precedes normalization/smoothing; stage means do not multiply to final p.

Paired effects against existing LAPA On | equal-protocol macro

Mean paired difference and 95% t interval (df=2); three training seeds, not new capture uncertainty.



Architecture, operator support and auxiliary objective also change; this is not an isolated causal test of removing QK.

Summary table | equal weight for each held-out protocol

Within-protocol field mean -> four-protocol mean -> three-seed mean; intervals are intentionally not clipped.

Model	p(target) [95% CI]	NLL [95% CI]	Hit@1 [95% CI]
RoPE / Off	0.3951 [0.2007, 0.5895]	5.5322 [3.2802, 7.7842]	0.4469 [0.1516, 0.7423]
RoPE / LAPA On	0.4208 [0.3458, 0.4959]	3.0632 [2.1700, 3.9565]	0.5345 [0.4604, 0.6086]
CoPE / Off	0.3470 [0.2846, 0.4093]	4.4288 [3.5998, 5.2578]	0.3739 [0.2664, 0.4814]
CoPE / LAPA On	0.4563 [0.3808, 0.5317]	3.5259 [2.4974, 4.5543]	0.5098 [0.5033, 0.5164]
TAPE / Off	0.3928 [0.2364, 0.5491]	7.6970 [4.5926, 10.8013]	0.4077 [0.3430, 0.4724]
TAPE / LAPA On	0.4360 [0.3072, 0.5648]	3.0353 [2.7300, 3.3406]	0.4874 [0.3385, 0.6364]
SDPA / Off	0.3811 [0.2277, 0.5345]	4.6658 [2.3815, 6.9502]	0.4469 [0.1567, 0.7372]
SDPA / LAPA On	0.4145 [0.2913, 0.5378]	3.5094 [3.2615, 3.7573]	0.4327 [0.2733, 0.5922]
Four-factor / Direct	0.2957 [0.0852, 0.5062]	3.8477 [3.3679, 4.3274]	0.3732 [0.1409, 0.6054]
Four-factor / Sink	0.3118 [0.2828, 0.3407]	3.4192 [3.2093, 3.6291]	0.4309 [0.2697, 0.5921]

Modbus and TLS targets are all END. High accuracy there alone does not establish successful length arithmetic.